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Reg. No.

Time : 3.00 hrs.

Quarterly Examination - 2025
CHEMISTRY

Max. Marks : 70

PART - I

Answer all questions. Choose the most appropriate answer from the given four alternatives and write the option for the corresponding answer. 15 x 1 = 15

1. Which metal is refined by Van Arkel method?
   a) Zr b) Tl c) Zr & Ti d) Ni
2. Which of the following is not sp^2 hybridised?
   a) Graphite b) Graphene c) Fullerene d) Dry ice
3. The basicity of hypophosphorous acid is
   a) 1 b) 2 c) 3 d) 4
4. The correct order of bond dissociation enthalpy of halogens in the following order is
   a) $Br_2 > I_2 < F_2 > C\ell_2$ b) $C\ell_2 > Br_2 > F_2 > I_2$ c) $F_2 > C\ell_2 > Br_2 > I_2$ d) $I_2 > Br_2 > C\ell_2 > F_2$
5. Which is used in smoke screen?
   a) Borax b) diborane c) phosphine d) potash alum
6. Which ion has the number of unpaired electron equivalent to V^{+3}.
   a) Ti^{+3} b) Fe^{+3} c) Ni^{+2} d) Cr^{+3}
7. The general electronic configuration of Lanthanides is
   a) $(Xe)4f^{0-14}$ b) $(Xe)\ 4f^{1-14}\ 5d^{0-1}\ 6s^2$ c) $(Xe)\ 5d^{0-1}$ d) $(Xe)\ 4f^{0-14}\ 5d^{1-10}\ 6s^2$
8. In a solid atom M occupies ccp lattice and (1/3) of tetrahedral voids are occupied atom N, Find the formula of solid formed by M and N.
   a) M_3N_2 b) M_3N c) MN_3 d) MN
9. In a first order reaction $X \rightarrow Y$, if k is the rate constant and the initial concentration of the reactant x is 0.1m then half life is
   a) $\left(\frac{\log 2}{k}\right)$ b) $\frac{0.693}{(0.1)\,k}$ c) $\left(\frac{\ell n 2}{k}\right)$ d) None of these
10. Predict the rate law of the following reaction $2A + 2B \rightarrow C + 2D$, if the concentration 'A' is doubled when 'B' is does not change the rate of reaction increased 4 times, and also when the contraction of 'B' is increases 2 times and concentration of 'A' is does not change, then the rate of reaction increased 2 times.
    a) rate = $k[A]\,[B]^2$ b) rate = $k[A]^2\,[B]$ c) rate = $k\,[A]\,[B]$ d) rate = $k\,[A]^{1/4}\,[B]2$
11. Conjugate base for Bronsted acids H_2O and HF are
    a) OH^- and H_2FH^+ respectively b) H_3O^+ and F^-, respectively c) OH^- and F^-, respectively
    d) H_3O^+ and H_2F^+, respectively
12. The relationship between the solubility product and molar solubility for $Ag_2(CrO_4)$ is
    a) Ksp = s3 b) Ksp = 4s3 c) ksp = s2 d) ksp = 3s2
13. In presence of diluted acids, when isopropyl benzene is oxidised by air, it gives
    a) C_6H_5–COOH b) C_6H_5 – $COCH_3$ c) C_6H_5–CO –C_6H_5 d) C_6H_5OH
14. Ethanoic acid $\xrightarrow{P/Br_2}$ 2- bromoethanoic acl. This reaction is called as.....
    a) Finkelstein reaction b) Haloform reaction c) Hell-Volhard-Zelinsky reaction d) none of these
15. Tollen's reagent is a) acidified $AgNO_3$ b) Ammonical $AgNO_3$(or) Ammonia with silver nitrate
    c) aqueous $AgNO_3$ d) Solid $AgNO_3$

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PART - II

Answer any six questions. Question number 24 is compulsory **6 x 2 = 12**

16. Write a note on gravity separation method. (or) hydraulic wash?
17. How is aluminium chloride prepared by MCAfee process?
18. Explain the bleaching action of chlorine.
19. Cu^{+2} ion compounds are coloured, but Zn^{+2} ion compounds are not coloured. why?
20. Differentiate octahedral voids from tetrahedral voids.
21. Identify the order of the following reaction.
    (i) Rusting of iron. (ii) Radioactive disintegration $_{92}U^{235}$
22. What is Bayer's reagent? By using this how will you convert ethene to Ethan-1, 2-diol.
23. Write a note on Haloform reaction.
24. Calculate the pH of 10^{-7}m HCl.

PART - III

Answer any six questions. Question number 33 is compulsory **6 x 3 = 18**

25. What is picric acid? How is it prepared?
26. Write the differences between Lewis and Lewis base.
27. (i) What is bragg equation? (ii) Classify the molecular crystal with suitable example.
28. Define half-life period of reaction. The half-life period of first order reaction is independent of initial concentration.
29. Explain magnetic separation method.
30. What is catenation? Write the catenation property of carbon.
31. (i) Why fluorine always exhibit - 1 oxidation state? Explain.
    (ii) Why HF cannot be stored in glass bottle? Explain.
32. What is lanthanide contraction? Write their consequences.
33. Compound 'A' of molecular formula C_3H_4, is react with H_2O, in present of Hg^{+2}/H_2SO_4 gives 'B'. 'B' undergoes haloform reaction. Compound 'B' also react with NH_2NH_2/C_2H_5ONa give hydrocarbon of compound 'C'. Identify compound A, B & C. Explain the reaction.

PART - IV

Answer all the following questions. **5 x 5 = 25**

34. a) (i) What are the differences between minerals and ores?
    (ii) What is role of silica on the extraction of copper?
    (iii) What is calcination? **(OR)**
    b) (i) Write the differences between graphite and diamond. (3)
    (ii) What is the action of heat on Borax. Write the equations.
35. a) (i) Explain the structure of Ammonia (ii) Mention the uses of Argon. **(OR)**
    b) (i) Write the preparation of potassium di-chromate.
    (ii) Write a note on Zeigler - Natta catalyst. Give its use.
36. a) (i) What is packing efficiency? Calculate the packing efficiency of body centred cubic crystal. **(OR)**
    b) (i) Derive integrated rate law for a first order reaction. A → product.
37. a) (i) Derive an expression for Ostwald's dilution law. **(OR)**
    b) Write a note on (i) Schatten-Baumann reaction (1½) (ii) Riemer - Tiemann reaction (2)
    (iii) Swern Oxidation (1½)
38. a) (i) Explain the mechanism of Cannizzaro reaction (3)
    (ii) What is transesterification? Give example. (2) **(OR)**
    b) (i) Explain common ion-effect with suitable example. (2) (ii) Compare lanthanides and actinides.

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